

How Plants Get the Energy to Grow How Plants Get the Energy to Grow

Name _____

Date _____

Two Pots, One Difference Two Pots, One Difference

Two pots are set up to be compared. They hold identical soil, they are watered by the same amount, and they stand in the same room at the same temperature. The only difference between them is **where each pot sits**. Two pots are set up to be compared. They hold identical soil, they are watered by the same amount, and they stand in the same room at the same temperature. The only difference between them is **where each pot sits**.

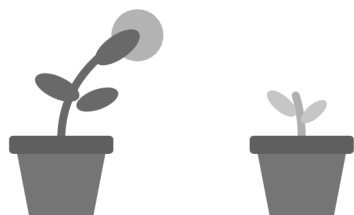


Figure 1: The two seedlings are treated the same in every way except where they stand. The two seedlings are treated the same in every way except where they stand.

A plant that adds height and unfolds new leaves is doing something that costs energy. That gives this lesson its question:

Where does the energy a plant uses to grow actually come from? A plant that adds height and unfolds new leaves is doing something that costs energy. That gives this lesson its question:

Where does the energy a plant uses to grow actually come from?

What you already know What you already know

- Plants are living things that need light and water to grow. Plants are living things that need light and water to grow.
- Every living thing needs energy to stay alive. Every living thing needs energy to stay alive.

Two possible answers Two possible answers

Where the growing energy comes from is exactly what people disagree about. Where the growing energy comes from is exactly what people disagree about.

ANSWER A ANSWER A

From the soil From the soil

The energy is taken in together with the water the roots draw up. The energy is taken in together with the water the roots draw up.

ANSWER B ANSWER B

From the light From the light

The energy is captured from the light by the plant's leaves. The energy is captured from the light by the plant's leaves.

Neither answer is settled yet. Take the one you think is more likely and hold it as a **prediction**, not a fact. Evidence, not preference, is what settles a question like this. Neither answer is settled yet. Take the one you think is more likely and hold it as a **prediction**, not a fact. Evidence, not preference, is what settles a question like this.

Q1.

Where does the energy a plant uses to grow come from? Where does the energy a plant uses to grow come from?

- ☐ **A.** The plant takes in its growing energy from the soil, together with the water. The plant takes in its growing energy from the soil, together with the water.
- ☐ **B.** The plant captures its growing energy from the light, using its leaves. The plant captures its growing energy from the light, using its leaves.

I think this because:

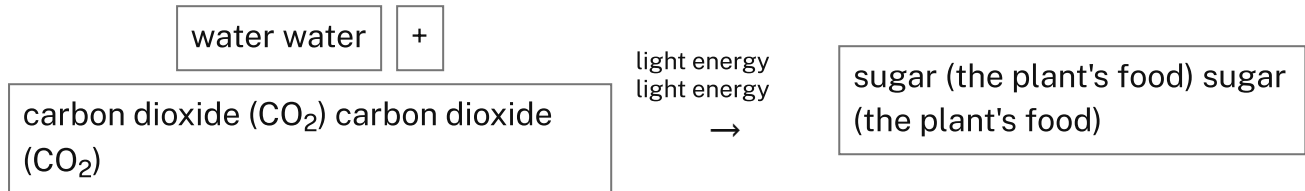
Where the Growing Energy Comes From Where the Growing Energy Comes From

Key idea Key idea

The energy a plant grows on is captured from light, not taken in from the soil. The energy a plant grows on is captured from light, not taken in from the soil.

The green parts of a plant — mostly its leaves — absorb **light energy**. The leaf uses that energy to make the plant's own **food**: a sugar built from water drawn up out of the soil and carbon dioxide taken in from the air. The green parts of a plant — mostly its leaves — absorb **light energy**. The leaf uses that energy to make the plant's own **food**: a sugar built from water drawn up out of the soil and carbon dioxide taken in from the air.

In one line In one line



The energy stored inside that sugar is what the plant spends on building new leaves, stem and roots. The energy stored inside that sugar is what the plant spends on building new leaves, stem and roots.

Follow the energy, step by step Follow the energy, step by step

1. Sunlight reaches the leaf. Sunlight reaches the leaf.
2. The leaf's green parts capture that light energy. The leaf's green parts capture that light energy.
3. The captured energy is used to build sugar from water and carbon dioxide. The energy is now stored inside food the plant has made for itself. The captured energy is used to build sugar from water and carbon dioxide. The energy is now stored inside food the plant has made for itself.
4. The plant breaks some of that sugar down again and uses the released energy to build new leaves, stem and roots. That is growth. The plant breaks some of that sugar down again and uses the released energy to build new leaves, stem and roots. That is growth.

Notice two things Notice two things

The energy enters the plant through the leaf, never through the soil. The only things arriving from the soil are water and dissolved minerals. Leaf area can be measured in m². The energy enters the plant through the leaf, never through the soil. The only things arriving from the soil are water and dissolved minerals. Leaf area can be measured in m².

So what is the soil for? So what is the soil for?

Soil does a real job. It holds the plant firmly in place, and it holds water and dissolved minerals, which the roots take in and which then travel up to the leaves. Minerals are raw materials. They are not food, and they carry no usable energy. Soil does a real job. It holds the plant firmly in place, and it holds water and dissolved minerals, which the roots take in and which then travel up to the leaves. Minerals are raw materials. They are not food, and they carry no usable energy.

WHAT THE PLANT TAKES IN WHAT THE PLANT TAKES IN	WHERE FROM WHERE FROM	IS IT ENERGY? IS IT ENERGY?
Water Water	Soil, through the roots Soil, through the roots	No. A raw material for making sugar. No. A raw material for making sugar.
Dissolved minerals Dissolved minerals	Soil, through the roots Soil, through the roots	No. Raw materials for building the plant's body. No. Raw materials for building the plant's body.
Light Light	Captured by the leaves Captured by the leaves	Yes. It ends up stored in sugar. Yes. It ends up stored in sugar.

So a plant standing in rich soil with plenty of water, but kept in the dark, cannot make its own food and cannot grow, however well watered it is. So a plant standing in rich soil with plenty of water, but kept in the dark, cannot make its own food and cannot grow, however well watered it is.

Two Beliefs That Fail the Evidence Two Beliefs That Fail the Evidence

Common belief 1 Common belief 1	
The belief	<i>"The soil feeds the plant. A plant kept in the dark will keep growing as long as it is watered. The soil feeds the plant. A plant kept in the dark will keep growing as long as it is watered."</i>
The evidence	Both pots held the same soil and got the same water. If the soil were supplying food and energy, the cupboard plant should have grown just as well. It stopped growing and turned pale. Both pots held the same soil and got the same water. If the soil were supplying food and energy, the cupboard plant should have grown just as well. It stopped growing and turned pale.
So	Soil supplies only water and dissolved minerals. The plant's food is made in its own leaves, using light energy. Soil supplies only water and dissolved minerals. The plant's food is made in its own leaves, using light energy.

Why did the cupboard plant keep going for a while? It was living on food it had already made and stored. Once that store ran down, it stopped growing and lost its green colour. Why did the cupboard plant keep going for a while? It was living on food it had already made and stored. Once that store ran down, it stopped growing and lost its green colour.

Common belief 2 Common belief 2	
The belief	<i>“Light only keeps a plant warm or makes it look healthy. The energy for growth comes from the water and the minerals. Light only keeps a plant warm or makes it look healthy. The energy for growth comes from the water and the minerals.”</i>
The evidence	Both pots stood in the same room at the same temperature. The cupboard was not cold, so the cupboard plant lacked no warmth. It still did not grow. Both pots stood in the same room at the same temperature. The cupboard was not cold, so the cupboard plant lacked no warmth. It still did not grow.
So	Light is not merely warmth. The leaves capture light energy and use it to make the plant's own food, and growth runs on the energy stored in that food. Light is not merely warmth. The leaves capture light energy and use it to make the plant's own food, and growth runs on the energy stored in that food.

Which Observation Challenges the Soil Idea? Which Observation Challenges the Soil Idea?

Plants get their food from the soil, so a plant in a dark room will keep growing as long as I water it. Plants get their food from the soil, so a plant in a dark room will keep growing as long as I water it.

A student says A student says

That claim is wrong. The strong move is not to repeat the explanation, but to point to **evidence the claim cannot explain**. That claim is wrong. The strong move is not to repeat the explanation, but to point to **evidence the claim cannot explain**.

Q2.

Which observation would best challenge the student's idea? Which observation would best challenge the student's idea?

- ☐ **A.** A watered plant kept in darkness becomes pale and stops growing larger. A watered plant kept in darkness becomes pale and stops growing larger.
- ☐ **B.** A watered plant in darkness grows taller because it stretches toward the light. A watered plant in darkness grows taller because it stretches toward the light.
- ☐ **C.** A watered plant in darkness stays cool, which proves it lacks growth energy. A watered plant in darkness stays cool, which proves it lacks growth energy.
- ☐ **D.** A watered plant in darkness grows well when it is given plant food. A watered plant in darkness grows well when it is given plant food.

Teacher copy — answer key

Q1. Prediction (not marked)

Not marked: accept either response.

Option	Teacher note
soil	The evidence will test this idea. The evidence will test this idea.

Q2. A

Feedback: That's the one. The soil idea says this plant should keep growing, because it still has soil and water. It does not, so the soil idea cannot be right. That's the one. The soil idea says this plant should keep growing, because it still has soil and water. It does not, so the soil idea cannot be right.

Option	Teacher note
b	A plant growing taller in the dark would fit the soil idea, not challenge it. A plant growing taller in the dark would fit the soil idea, not challenge it.
c	Temperature is not the issue. In the two-pot comparison both plants were equally warm. Temperature is not the issue. In the two-pot comparison both plants were equally warm.
d	Growing well on plant food in the dark would support the soil idea, not challenge it. Growing well on plant food in the dark would support the soil idea, not challenge it.